



Data Scientist

QP Code: SSC/Q8104

Version: 4.0

NSQF Level: 6

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Qualification Pack

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SSC/Q8104: Data Scientist

Brief Job Description

A data scientist is a skilled professional responsible for collecting, cleaning, analyzing, and interpreting large volumes of data to derive actionable insights that support business decision-making. They employ statistical methods, machine learning algorithms, and various data processing tools to explore data, build predictive models, and validate results. Data scientists also develop scalable data pipelines, ensure data quality, and create visualizations and dashboards to communicate findings effectively. Their role spans across understanding business requirements, designing data strategies, implementing algorithms, deploying models into production environments, and continually monitoring and refining those models to adapt to changing data patterns. Overall, a data scientist bridges the gap between raw data and strategic insights, empowering organizations to make informed, data-driven decisions.

Personal Attributes

A data scientist needs to have excellent analytical skills, attention to detail, critical thinking and problem solving ability. S/he needs to have strong communication skills and a superior understanding of the business to work with stakeholders and decision makers across the organization.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SSC/N8157: Collect & process data and ensure data quality standards are achieved](#)
2. [SSC/N8103: Perform exploratory and Inferential analysis of the data](#)
3. [SSC/N8104: Perform research, design & validation of algorithmic models](#)
4. [SSC/N8106: Risk Analysis & Mitigation for Algorithmic Models](#)
5. [SSC/N8108: Define business outcomes and create visualizations from results of the analysis](#)
6. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Artificial Intelligence and Big Data Analytics

Qualification Pack

Country	India
NSQF Level	6
Credits	21
Aligned to NCO/ISCO/ISIC Code	NCO-2015/NIL
Minimum Educational Qualification & Experience	Completed 4 year UG program (** PHD or PG or UG or diploma with courses related to Engg./ Science) OR Completed 3 year UG degree (PHD or PG or UG or diploma with courses related to Engg./ Science) with 1.5 years of experience Relevant Experience in job roles related to IT/AI/Big Data Analytics/Programming The relevant experience would include work, internship, and apprenticeship after completing relevant educational qualifications OR Previous relevant Qualification of NSQF Level (5.5) with 1.5 years of experience Relevant Experience in job roles related to IT/AI/Big Data Analytics/Programming The relevant experience would include work, internship, and apprenticeship after completing relevant educational qualifications
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	24 Years
Last Reviewed On	NA
Next Review Date	07/10/2028
NSQC Approval Date	07/10/2025
Version	4.0
Reference code on NQR	QG-06-IT-045952025-V2-NASSCOM
NQR Version	2

Qualification Pack

SSC/N8157: Collect & process data and ensure data quality standards are achieved

Description

Collect & process data and ensure data quality standards are achieved

Scope

The scope covers the following :

- Establish Data Quality Criteria and Thresholds, Develop Processes and Controls for Data Quality Assurance, Design and Maintain Data Schemas and Validation Rules, Implement Data Loading, Merging, and Real-time Data Streaming, Set Up Monitoring, Alerts, and Dashboards for Data Quality, Conduct Regular Data Audits and Performance Reviews, Establish Feedback Loops and Continuous Improvement Processes, Promote Cross-functional Collaboration and Training, Maintain Comprehensive Documentation, Foster a Culture of Data Quality and Use Emerging Tools

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Determine clear criteria for data quality dimensions (accuracy, completeness, consistency, timeliness, validity, relevancy) and acceptable benchmarks, tailored to organizational needs
- PC2.** Create and implement processes, automated scripts, validation rules, and controls to regularly assess data against the defined standards.
- PC3.** Define data structures aligning with organizational databases, and encode rules to validate incoming data to ensure it meets quality criteria before entry.
- PC4.** Transfer validated data into data warehouses or databases, ensuring seamless merging with existing datasets, and employ real-time solutions where applicable
- PC5.** Use tools to automate monitoring, generate alerts, and visualize data quality metrics in real time for quick detection and response to issues
- PC6.** Schedule and perform audits to validate data compliance, analyze discrepancies, and perform gap analysis to identify areas for improvement
- PC7.** Gather insights from audits, monitoring, and end-users to refine standards, processes, and controls continually
- PC8.** Work closely with IT, data science, and business teams to align efforts and develop training programs to foster a culture of data quality
- PC9.** Keep detailed records of data quality standards, procedures, incidents, and improvement actions for transparency and audit readiness
- PC10.** Encourage a proactive approach toward continuous improvement, leveraging new technologies, and fostering staff awareness of data quality principles

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** the purpose and aims of the statistical analysis being undertaken
- KU2.** organizational policies, procedures and guidelines which relate to preprocessing data
- KU3.** different data sources and how to access documents and information from data sources
- KU4.** organizational policies and procedures for sharing data
- KU5.** whom to consult while preprocessing data
- KU6.** the range of standard templates and tools available and how to use them
- KU7.** the difference between various types of data. For example, qualitative vs quantitative data, processed vs unprocessed data, discrete vs continuous data
- KU8.** different statistical analysis software, packages, libraries and tools that can be used to preprocess data such as R or Pandas
- KU9.** different functions to identify and remove missing values
- KU10.** different functions to identify and transform data types of variables such as integer, float, character
- KU11.** different methodological approaches for normalizing the dataset such as standard score, feature scaling, etc.
- KU12.** different data formats and structures
- KU13.** how to index and organize data
- KU14.** how to identify and refer anomalies in data
- KU15.** how to work on various databases and operating systems

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** impact analysis of the various actions performed and disseminating relevant information to others
- GS2.** analyze data and understand its implications on business
- GS3.** check your work is complete and free from errors

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Determine clear criteria for data quality dimensions (accuracy, completeness, consistency, timeliness, validity, relevancy) and acceptable benchmarks, tailored to organizational needs	3	5	-	2
PC2. Create and implement processes, automated scripts, validation rules, and controls to regularly assess data against the defined standards.	3	5	-	2
PC3. Define data structures aligning with organizational databases, and encode rules to validate incoming data to ensure it meets quality criteria before entry.	3	5	-	2
PC4. Transfer validated data into data warehouses or databases, ensuring seamless merging with existing datasets, and employ real-time solutions where applicable	3	5	-	2
PC5. Use tools to automate monitoring, generate alerts, and visualize data quality metrics in real time for quick detection and response to issues	3	5	-	2
PC6. Schedule and perform audits to validate data compliance, analyze discrepancies, and perform gap analysis to identify areas for improvement	3	5	-	2
PC7. Gather insights from audits, monitoring, and end-users to refine standards, processes, and controls continually	3	5	-	2
PC8. Work closely with IT, data science, and business teams to align efforts and develop training programs to foster a culture of data quality	3	5	-	2
PC9. Keep detailed records of data quality standards, procedures, incidents, and improvement actions for transparency and audit readiness	3	5	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. Encourage a proactive approach toward continuous improvement, leveraging new technologies, and fostering staff awareness of data quality principles	3	5	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8157
NOS Name	Collect & process data and ensure data quality standards are achieved
Sector	IT-ITeS
Sub-Sector	
Occupation	Artificial Intelligence and Big Data Analytics
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8103: Perform exploratory and Inferential analysis of the data

Description

This unit is about using a variety of techniques to perform exploratory analysis to describe and summarize data for internal and external clients

Scope

The scope covers the following :

- Define the dataset
- Summarize and optimize the dataset

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Identify the data types for each variable of the dataset
- PC2.** Identify the key variables required for modelling or analysis
- PC3.** Use statistical techniques to summarize the key variables in the dataset
- PC4.** Describe summary statistics for key variables using graphical formats
- PC5.** Perform dimension reduction to optimize the variables in the dataset, if required
- PC6.** Define the correlation factors using clustering and other techniques
- PC7.** Validate data using appropriate tools and processes
- PC8.** Repeat the analysis iteratively to arrive at optimal results
- PC9.** Validate the final output in consultation with the relevant stakeholders
- PC10.** Gain inferences from the final output of the data analysis
- PC11.** Develop a hypothesis model to explain the discovered inferences
- PC12.** Evaluate the results of the analysis and define business outcomes
- PC13.** Define prescriptive actions based on the defined business outcomes

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the purpose and aims of the statistical analysis being undertaken
- KU2.** organizational policies, procedures and guidelines which relate to performing exploratory analysis
- KU3.** different data sources and how to access documents and information from data sources
- KU4.** organizational policies and procedures for sharing data
- KU5.** who to involve when performing exploratory analysis
- KU6.** the range of standard templates and tools available and how to use them
- KU7.** the difference between various types of data. For example, qualitative vs quantitative data processed vs unprocessed data discrete vs continuous data

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- KU8.** different statistical analysis software, packages, libraries and tools that can be used to preprocess data such as R or Pandas
- KU9.** different statistical analysis software, packages, libraries and tools that can be used to summarize data such as R, Numpy, Statsmodels, or Pandas
- KU10.** different functions to summarize variables across different data types such as integer, float, or character
- KU11.** different graphical formats to describe summary statistics
- KU12.** different methodological approaches for dimension reduction such as PCA, LDA, or NMF
- KU13.** different methodological approaches for defining correlations between variables such as the scatter diagram method, correlation coefficients, method of least squares
- KU14.** multivariate visualizations, for mapping and understanding interactions between different fields in the data
- KU15.** how to make inferences from analysed data and explain it using a hypothesis model
- KU16.** different types of prescriptive actions
- KU17.** how to identify and refer anomalies in data
- KU18.** how to work on various operating systems such as linux, ubuntu, or windows

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** impact analysis of the various actions performed and disseminating relevant information to others
- GS2.** analyze data and understand its implications on business

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Identify the data types for each variable of the dataset	2	3	-	1
PC2. Identify the key variables required for modelling or analysis	2	4	-	2
PC3. Use statistical techniques to summarize the key variables in the dataset	2	7	-	2
PC4. Describe summary statistics for key variables using graphical formats	2	3	-	1
PC5. Perform dimension reduction to optimize the variables in the dataset, if required	2	3	-	1
PC6. Define the correlation factors using clustering and other techniques	3	4	-	2
PC7. Validate data using appropriate tools and processes	3	4	-	2
PC8. Repeat the analysis iteratively to arrive at optimal results	2	3	-	1
PC9. Validate the final output in consultation with the relevant stakeholders	2	3	-	1
PC10. Gain inferences from the final output of the data analysis	2	4	-	2
PC11. Develop a hypothesis model to explain the discovered inferences	2	4	-	2
PC12. Evaluate the results of the analysis and define business outcomes	3	4	-	2
PC13. Define prescriptive actions based on the defined business outcomes	3	4	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8103
NOS Name	Perform exploratory and Inferential analysis of the data
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Artificial Intelligence and Big Data Analytics
NSQF Level	5
Credits	4
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

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SSC/N8104: Perform research, design & validation of algorithmic models

Description

This unit is about performing research and designing a variety of algorithmic models for internal and external clients.

Scope

The scope covers the following :

- Define hypothesis
- Select model
- Prototype and design

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Identify and clearly define the analysis objectives based on business needs
- PC2.** Develop hypotheses and select appropriate libraries, frameworks, and tools to address the objectives.
- PC3.** Conduct research on existing statistical and machine learning models for suitability
- PC4.** Choose the mode of learning (supervised, unsupervised, reinforcement) based on data and objectives
- PC5.** Prototype and develop the algorithmic models aligned with project requirements
- PC6.** Validate models using proper testing techniques, ensuring accuracy, robustness, and fairness.
- PC7.** Optimize models through hyperparameter tuning and iterative improvements
- PC8.** Address issues like overfitting, underfitting, residual errors, and dispersion errors.
- PC9.** Document model design, assumptions, and versioning for future reference
- PC10.** Set success metrics for model performance and evaluate models against these metrics
- PC11.** Establish data flows and manage data inputs needed for model training and testing
- PC12.** Implement continuous retraining where applicable to maintain model relevance

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the purpose and aims of the analysis being undertaken
- KU2.** organizational policies and procedures for sharing data
- KU3.** who to involve when managing administration of production systems and operations
- KU4.** different continuous delivery and deployment models
- KU5.** how to design and develop a staging environment
- KU6.** infrastructure as code processes

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- KU7.** different configuration management tools such as Puppet, Chef
- KU8.** different continuous integration tools such as Jenkins, Travis CI, Bamboo
- KU9.** different application release automation tools such as BuildMaster, Puppet, FlexDeploy
- KU10.** different build automation tools such as Gradle, Jenkins
- KU11.** different application lifecycle management tools such as GitLab, JIRA
- KU12.** different methodological approaches to testing such as unit and integration testing
- KU13.** different test and QA automation tools such as selenium
- KU14.** how to work on various operating systems such as linux, ubuntu, or windows

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** make decisions on suitable courses
- GS2.** plan and organize your own work to achieve targets and deadlines
- GS3.** refer anomalies to the supervisor
- GS4.** check your work is complete and free from errors

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Identify and clearly define the analysis objectives based on business needs	2	2	-	1
PC2. Develop hypotheses and select appropriate libraries, frameworks, and tools to address the objectives.	2	2	-	1
PC3. Conduct research on existing statistical and machine learning models for suitability	2	3	-	1
PC4. Choose the mode of learning (supervised, unsupervised, reinforcement) based on data and objectives	2	4	-	1
PC5. Prototype and develop the algorithmic models aligned with project requirements	2	4	-	2
PC6. Validate models using proper testing techniques, ensuring accuracy, robustness, and fairness.	2	4	-	2
PC7. Optimize models through hyperparameter tuning and iterative improvements	3	5	-	2
PC8. Address issues like overfitting, underfitting, residual errors, and dispersion errors.	3	5	-	2
PC9. Document model design, assumptions, and versioning for future reference	3	5	-	2
PC10. Set success metrics for model performance and evaluate models against these metrics	3	5	-	2
PC11. Establish data flows and manage data inputs needed for model training and testing	3	6	-	2
PC12. Implement continuous retraining where applicable to maintain model relevance	3	5	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8104
NOS Name	Perform research, design & validation of algorithmic models
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Artificial Intelligence and Big Data Analytics, Artificial Intelligence and Big Data Analytics
NSQF Level	5
Credits	5
Version	4.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

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SSC/N8106: Risk Analysis & Mitigation for Algorithmic Models

Description

This unit is about evaluating the risks in deploying algorithmic models and developing mitigation measures for internal and external clients

Scope

The scope covers the following :

- Define model scope
- Analyse algorithmic model

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Define the objectives, expected outcomes, and applicable performance metrics
- PC2.** Identify data sources, assumptions, and potential biases in model design
- PC3.** Test models across different inputs/scenarios to understand variability and risks
- PC4.** Quantify potential risks such as unexpected outcomes, bias, or model failures
- PC5.** Develop mitigation strategies, checks, and validation processes to minimize risks
- PC6.** Document risk assessments, mitigation measures, and validation outcomes
- PC7.** Engage stakeholders to validate risk management plans
- PC8.** Implement corrective actions and iteratively improve risk controls

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the purpose and aims of the analysis being undertaken
- KU2.** organizational policies, procedures and guidelines which relate to evaluating risks associated with algorithmic models
- KU3.** different data sources and how to access documents and information from data sources
- KU4.** organizational policies and procedures for sharing data
- KU5.** who to involve when evaluating business performance of algorithmic models
- KU6.** the range of standard templates and tools available and how to use them
- KU7.** different performance metrics to monitor business outcomes of algorithmic models
- KU8.** different methodological approaches for identifying model hyperparameters such as grid search, random search, bayesian optimization
- KU9.** how to tune hyperparameter configurations
- KU10.** how to identify and refer anomalies in data
- KU11.** how to work on various operating systems such as linux, ubuntu, or windows

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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** impact analysis of the various actions performed and disseminating relevant information to others
- GS2.** check your work is complete and free from errors

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Define the objectives, expected outcomes, and applicable performance metrics	3	6	-	2
PC2. Identify data sources, assumptions, and potential biases in model design	3	6	-	2
PC3. Test models across different inputs/scenarios to understand variability and risks	3	6	-	2
PC4. Quantify potential risks such as unexpected outcomes, bias, or model failures	3	6	-	2
PC5. Develop mitigation strategies, checks, and validation processes to minimize risks	3	6	-	2
PC6. Document risk assessments, mitigation measures, and validation outcomes	7	7	-	4
PC7. Engage stakeholders to validate risk management plans	5	7	-	4
PC8. Implement corrective actions and iteratively improve risk controls	3	6	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8106
NOS Name	Risk Analysis & Mitigation for Algorithmic Models
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Artificial Intelligence & Big Data Analytics
NSQF Level	6
Credits	2
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

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SSC/N8108: Define business outcomes and create visualizations from results of the analysis

Description

This unit is about defining business outcomes from results of a statistical analysis and create visualizations to report them.

Scope

The scope covers the following :

- Define the scope
- Report business outcomes

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Identify the objective of the analysis
- PC2.** Establish the purpose, scope, and target audience to report the business outcomes
- PC3.** Define the delivery mode and format such as excel sheets, reports, APIs to report the business outcomes
- PC4.** Summarize the defined business outcomes into a narrative
- PC5.** Select suitable visualizations to represent the defined business outcomes
- PC6.** Present outcomes through selected visualizations using standard templates and agreed language standards
- PC7.** Validate visualizations with appropriate people
- PC8.** Publish visualizations for consumption across all agreed formats

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the purpose and aims of the statistical analysis being undertaken
- KU2.** organizational policies, procedures and guidelines which relate to creating visualizations
- KU3.** different data sources and how to access documents and information from data sources
- KU4.** organizational policies and procedures for sharing data
- KU5.** who to involve when defining business outcomes and creating visualizations
- KU6.** intended audiences for reporting business outcomes
- KU7.** their organization's knowledge base and how to access and update this
- KU8.** organizational processes and procedures for approving and publishing documents
- KU9.** the range of standard templates and tools available and how to use these
- KU10.** different statistical concepts such as distributions, hypothesis testing, confidence intervals etc.

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- KU11.** how to evaluate results of quantitative or qualitative analysis to define business outcomes
- KU12.** different graphical formats for presenting data and how to create these
- KU13.** different styles used in visualizations, including your organizations house style, types & templates
- KU14.** different applications, libraries or packages to create visualizations using tools such as Tableau, Qlikview, d3js etc
- KU15.** different change management procedures, including version control and approvals
- KU16.** how to identify and refer anomalies in data
- KU17.** how to work on various operating systems such as linux, ubuntu, or windows

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete accurate well written work with attention to detail
- GS2.** follow instructions, guidelines, procedures, rules and service level agreements
- GS3.** check that your own and/or your peers work meets customer requirements
- GS4.** work effectively in a customer facing environment
- GS5.** analyze data and activities
- GS6.** ability of understand business impact and disseminate relevant information to others
- GS7.** check your work is complete and free from errors

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Identify the objective of the analysis	4	6	-	2
PC2. Establish the purpose, scope, and target audience to report the business outcomes	4	6	-	3
PC3. Define the delivery mode and format such as excel sheets, reports, APIs to report the business outcomes	4	6	-	3
PC4. Summarize the defined business outcomes into a narrative	4	7	-	3
PC5. Select suitable visualizations to represent the defined business outcomes	4	7	-	3
PC6. Present outcomes through selected visualizations using standard templates and agreed language standards	4	6	-	2
PC7. Validate visualizations with appropriate people	3	6	-	2
PC8. Publish visualizations for consumption across all agreed formats	3	6	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8108
NOS Name	Define business outcomes and create visualizations from results of the analysis
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Artificial Intelligence and Big Data Analytics
NSQF Level	5
Credits	4
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

The assessment will consist of a blend of hands-on practical evaluations, viva-voce, and online proctored scenario-based multiple-choice questions ensuring a thorough evaluation of the individual's proficiency in learning outcomes, practical understanding, and real-world application of concepts.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N8157.Collect & process data and ensure data quality standards are achieved	30	50	0	20	100	18
SSC/N8103.Perform exploratory and Inferential analysis of the data	30	50	0	20	100	18
SSC/N8104.Perform research, design & validation of algorithmic models	30	50	0	20	100	18
SSC/N8106.Risk Analysis & Mitigation for Algorithmic Models	30	50	0	20	100	18
SSC/N8108.Define business outcomes and create visualizations from results of the analysis	30	50	0	20	100	18
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	170	280	-	100	550	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.